

Can Machine Learning Reduce the Computational Cost of Monte Carlo Tail-Risk Estimation Without Sacrificing Accuracy?

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Abstract

Monte Carlo full revaluation is flexible for estimating the risk of nonlinear portfolios but can become computationally expensive when many instruments must be repriced across large scenario sets. This paper investigates whether machine-learning surrogate models can approximate portfolio profit and loss sufficiently accurately to reduce this computational cost while preserving the quality of extreme-tail risk estimates. A synthetic portfolio of 500 European options across five equities was constructed and approximately delta-neutralised using stock positions. Ten-day scenarios were generated from a correlated, variance-standardised multivariate Student- t model with eleven risk factors: five equity returns, five volatility shocks and one interest-rate shock. Full revaluation was implemented in C++, while Histogram Gradient Boosting and multilayer perceptron surrogates were trained in Python. The multilayer perceptron substantially outperformed the tree-based model. Across five independent training samples, a 50,000-observation MLP achieved median absolute errors of 0.69% for 99% VaR and 3.72% for 99% Expected Shortfall, with 89.37% overlap with the true worst 1% of scenarios. Its recurring one-million-scenario computation required approximately 1.20 seconds compared with 30.01 seconds for direct full revaluation, a compute-only speedup of roughly 25 times. The results suggest that machine-learning surrogates can materially reduce Monte Carlo tail-risk computation, although residual tail bias remains and must be validated explicitly.

1 Introduction

Monte Carlo simulation is an important tool in derivatives pricing and financial risk management because it allows portfolio values to be evaluated under large numbers of possible market scenarios (Glasserman, 2004). Its flexibility is particularly useful for nonlinear portfolios, but its main disadvantage is computational cost: a large derivatives portfolio may require hundreds of thousands or millions of scenarios, with every instrument repriced under each scenario. Expected Shortfall has also become particularly relevant to modern market-risk measurement; for example, the Basel market-risk framework uses Expected Shortfall in its internal-model approach rather than relying exclusively on Value-at-Risk (Basel Committee on Banking Supervision, 2019). Although the present study is not intended to reproduce the Basel regulatory framework, this provides practical motivation for accurately estimating losses in the extreme tail.

A possible way to reduce repeated revaluation cost is to train a machine-learning surrogate that learns the relationship between market risk factors and portfolio value or P&L. Previous research has demonstrated that supervised-learning methods can be used for portfolio pricing and risk calculations (Fernandez-Arjona and Filipović, 2022), while neural-network surrogates have also been shown to accelerate computationally demanding option-pricing models (Chen et al., 2026). This paper investigates the problem from a different angle by comparing tree-based and neural-network surrogates and explicitly studying how the amount of expensive

full-revaluation training data affects both extreme-tail accuracy and computational break-even. The research question is therefore: *how much full-revaluation training data is required for a machine-learning surrogate to estimate extreme portfolio tail risk accurately, and when does the resulting computational saving outweigh the surrogate error?*

2 Methodology

2.1 Portfolio and Pricing

The synthetic portfolio contains 500 European options across five equities. Initial equity prices are (100, 80, 120, 150, 60), initial implied volatilities are (0.18, 0.22, 0.25, 0.30, 0.35), dividend yields are (0.01, 0.015, 0.005, 0.02, 0) and the initial continuously compounded risk-free rate is 4%. For each equity, call and put options are created across ten strike multipliers from 0.75 to 1.25 and five maturities from 0.25 to 1.50 years, producing $5 \times 10 \times 5 \times 2 = 500$ option positions. Long and short quantities alternate deterministically across the grid. Stock positions are then added to approximately neutralise the initial equity delta of each underlying, reducing simple first-order directional exposure and increasing the relative importance of nonlinear option effects.

European options are valued using the Black–Scholes–Merton framework (Black and Scholes, 1973; Merton, 1973). For a European call option, the value is

$$C = S e^{-qT} N(d_1) - K e^{-rT} N(d_2). \quad (1)$$

Here, C is the call value, S is the current underlying price, K is the strike price, T is the remaining time to maturity, r is the continuously compounded risk-free rate, q is the continuous dividend yield and $N(\cdot)$ is the standard normal cumulative distribution function. The first intermediate quantity is

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2) T}{\sigma\sqrt{T}}. \quad (2)$$

In this expression, σ is the implied volatility of the underlying. The numerator combines log-moneyness with the effect of rates, dividends and volatility over the remaining life of the option, while the denominator scales this quantity by volatility and time. The second intermediate quantity is $d_2 = d_1 - \sigma\sqrt{T}$. The difference between d_1 and d_2 therefore reflects the additional uncertainty associated with volatility over the remaining maturity. European put options are valued using

$$P = K e^{-rT} N(-d_2) - S e^{-qT} N(-d_1), \quad (3)$$

where P denotes the put value and the remaining quantities are defined as above.

The portfolio is approximately delta-neutralised separately for each underlying. If $\mathcal{O}_i$ is the set of options written on underlying i , Q_j is the quantity of option j and Δ_j is its Black–Scholes delta, the stock hedge is chosen according to

$$H_i = - \sum_{j \in \mathcal{O}_i} Q_j \Delta_j. \quad (4)$$

Thus the initial first-order exposure from the options is offset by an opposing stock position. The pricing engine, portfolio representation, scenario generator and full-revaluation Monte Carlo engine were implemented in C++20, and the final implementation passed eleven automated C++ tests covering the principal components of the system.

2.2 Scenario Generation

The simulation horizon is ten trading days, or $10/252$ years. Each scenario contains eleven risk factors: five equity log returns, five absolute implied-volatility shocks and one absolute interest-rate shock. A ten-day horizon is also used as a base horizon in the Basel market-risk framework, although the scenario system developed here is a synthetic research model rather than a regulatory implementation (Basel Committee on Banking Supervision, 2019). Empirical financial returns exhibit heavy tails and extreme movements that are poorly represented by a simple Gaussian model (Cont, 2001); consequently, the scenarios are generated using a variance-standardised multivariate Student- t construction with six degrees of freedom.

The multivariate Student- t distribution can be represented through a Gaussian vector multiplied by a common random scale (Demarta and McNeil, 2005). Let $\mathbf{Z}$ denote the correlated Gaussian factor vector and let U be chi-squared distributed with $\nu = 6$ degrees of freedom. The variance-standardising scale used in the simulation is

$$A = \sqrt{\frac{\nu - 2}{U}}. \quad (5)$$

Multiplying the Gaussian factor vector by this common scale gives

$$\mathbf{X} = A\mathbf{Z}. \quad (6)$$

The factor $\sqrt{(\nu - 2)/U}$ is used so that the marginal factors have unit variance before the desired shock scales are applied. This allows the specified spot, volatility and rate scales to retain a standard-deviation interpretation while introducing common heavy-tail dependence across the scenario.

The synthetic dependence assumptions are a spot-spot correlation of 0.35, a volatility-volatility correlation of 0.50 and a spot-volatility correlation of -0.30 , while the rate factor has zero Gaussian correlation with the remaining factors. The negative spot-volatility relationship is a modelling assumption motivated by the well-documented inverse relationship between equity values and equity volatility (Christie, 1982). These values are not estimated from market data and should therefore be viewed as controlled experimental assumptions rather than empirical calibration.

For each scenario, the simulated equity log return is applied as $S'_i = S_i e^{R_i}$. Volatility is updated as $\sigma'_i = \max(\sigma_i + \Delta\sigma_i, 0.0001)$, ensuring that extreme negative shocks cannot produce a non-positive volatility. The shocked interest rate is $r' = r + \Delta r$. Each option is then fully repriced at the ten-day horizon and portfolio P&L is calculated as $\text{P\&L}_s = V_s - V_0$, where V_0 is calculated once at inception and V_s is the portfolio value under scenario s .

2.3 Tail-Risk Measures

Value-at-Risk is a quantile-based measure of the loss distribution. Defining portfolio loss as $L = -\text{P\&L}$, VaR at confidence level α is

$$\text{VaR}_\alpha = \inf \{l : \Pr(L \leq l) \geq \alpha\}. \quad (7)$$

It therefore gives the loss threshold associated with a chosen confidence level, but does not describe the magnitude of losses once that threshold has been exceeded. Quantile-based measures such as VaR can also lack desirable properties such as subadditivity in general settings, motivating the development of coherent risk measures (Artzner et al., 1999).

Expected Shortfall instead measures losses within the tail itself. For a continuous loss distribution it can be written as

$$\text{ES}_\alpha = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha]. \quad (8)$$

Expected Shortfall, also closely related to Conditional Value-at-Risk, explicitly captures the severity of losses beyond the VaR threshold and has important theoretical advantages for financial risk measurement (Rockafellar and Uryasev, 2002). The primary measures in this experiment are 99% VaR and 99% ES. On the one-million-scenario benchmark, 99% ES therefore corresponds to the average loss among the worst 10,000 scenarios. A 97.5% ES measure is also recorded as a secondary robustness metric; the latter confidence level corresponds to that used by the Basel internal-model framework (Basel Committee on Banking Supervision, 2019).

2.4 Datasets

A 50,000-scenario training set, a separate 50,000-scenario validation set and a one-million-scenario benchmark set were generated using the C++ full-revaluation engine. The validation sample was used during model development, while the larger benchmark provides a fixed high-precision scenario set for comparison of the surrogate and full-revaluation methods.

To assess whether conclusions were dependent on one favourable training sample, five independent 250,000-observation training pools were also generated. Nested samples of 2,500, 5,000, 10,000, 20,000, 50,000, 100,000 and 250,000 observations were extracted from each pool. The same frozen model specification was therefore fitted across 35 training datasets, allowing both the effect of training size and variation between independent training samples to be examined.

2.5 Machine-Learning Models

The first surrogate was Histogram Gradient Boosting (HGB), a histogram-based implementation of gradient-boosted regression trees. Gradient boosting constructs an additive prediction model by sequentially fitting new learners to reduce the remaining loss, with regression trees forming a widely used base learner (Friedman, 2001). Initial experiments showed that reasonable overall predictive accuracy did not necessarily imply good extreme-tail performance, so tail-weighted HGB variants were investigated. The final HGB specification gave greater training weight to observations belonging to the worst 5% and worst 1% of the training loss distribution.

The second surrogate was a feed-forward multilayer perceptron (MLP), making it a natural candidate for approximating the smooth but nonlinear relationship between market risk factors and Black–Scholes portfolio P&L. The final network has architecture $11 \rightarrow 64 \rightarrow 64 \rightarrow 32 \rightarrow 1$. Hidden layers use rectified linear unit activations, while optimisation uses Adam. The model uses a batch size of 256, an initial learning rate of 10^{-3} , L2 regularisation with $\alpha = 10^{-3}$, a maximum of 300 iterations and early stopping with a 10% internal validation fraction.

Both the eleven input features and the P&L target are standardised using statistics calculated from the training sample. A small validation-based tuning exercise compared alternative network widths, activation functions, learning rates and regularisation strengths before the final MLP specification was frozen. The machine-learning models and preprocessing routines were implemented using scikit-learn.

2.6 Evaluation Metrics

Overall prediction quality is evaluated using root mean squared error, mean absolute error and R^2 , while tail-risk accuracy is evaluated using absolute percentage errors in 99% VaR and 99% ES. These scalar risk measures alone do not reveal whether the surrogate identifies the correct extreme scenarios, so additional tail-specific diagnostics are included. RMSE is calculated separately on the true worst 5% and worst 1% of full-revaluation losses, and mean loss error is also measured within the true worst 1%.

Tail overlap measures whether the surrogate identifies the same scenarios as extreme as the full-revaluation engine. If $\mathcal{T}$ denotes the set containing the true worst 1% of scenarios and $\hat{\mathcal{T}}$ denotes the surrogate’s predicted worst 1%, then

$$\text{Tail Overlap} = 100 \times \frac{|\mathcal{T} \cap \hat{\mathcal{T}}|}{|\mathcal{T}|}. \quad (9)$$

An overlap of 100% would therefore mean that the surrogate identifies exactly the same scenarios as belonging to the worst 1% of the loss distribution.

3 Results

3.1 MLP Performance and Training-Set Size

The one-million-scenario full-revaluation benchmark produced a 99% VaR of 2.1231 and a 99% ES of 3.5478. The MLP showed a clear improvement as the quantity of full-revaluation training data increased. Table 1 reports median performance across the five independent training samples.

Table 1: Median MLP performance across five independent training samples.

Labels	R^2	VaR Error (%)	ES Error (%)	Tail Overlap (%)
2,500	0.690	8.59	27.07	69.26
5,000	0.758	3.89	18.57	77.29
10,000	0.796	1.31	11.58	80.09
20,000	0.841	2.34	10.88	83.90
50,000	0.909	0.69	3.72	89.37
100,000	0.946	1.16	2.83	91.22
250,000	0.965	0.75	2.33	91.98

The results show a strong learning curve. Median R^2 rises from 0.690 with 2,500 observations to 0.965 with 250,000 observations, while tail overlap increases from 69.26% to 91.98%. The largest improvement in practical tail-risk accuracy occurs between 20,000 and 50,000 observations, where median 99% ES error falls from 10.88% to 3.72%. Increasing the training set beyond 50,000 observations continues to improve the underlying surrogate, particularly according to R^2 , worst-tail RMSE and tail overlap, although the gains in the scalar VaR and ES estimates become smaller and are not perfectly monotonic.

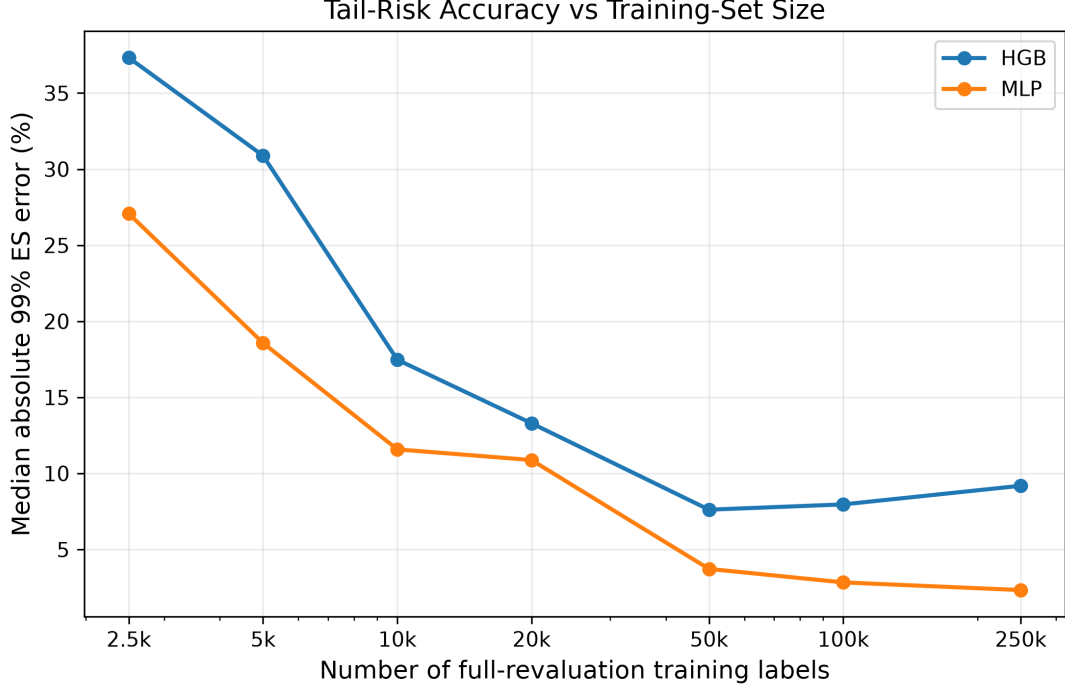


Figure 1: Median absolute 99% Expected Shortfall error against training-set size.

3.2 HGB versus MLP

The neural network consistently outperformed HGB. At 50,000 observations, HGB had median $R^2 = 0.660$, median 99% ES error of 7.62% and tail overlap of 74.92%, whereas the corresponding MLP achieved $R^2 = 0.909$, ES error of 3.72% and tail overlap of 89.37%. The difference remained substantial at 250,000 observations: HGB achieved median $R^2 = 0.701$ and ES error of 9.19%, compared with $R^2 = 0.965$ and ES error of 2.33% for the MLP. The improvement therefore cannot be explained simply by increasing the amount of training data; surrogate architecture itself materially affects the quality of the approximation.

One plausible explanation is the nature of the target function. Black–Scholes portfolio P&L is a relatively smooth nonlinear function of the underlying market variables, whereas tree ensembles approximate functions through collections of piecewise-constant regions. The feed-forward network is therefore well suited to learning a continuous pricing surface, consistent with the broader use of neural networks as financial valuation surrogates (Fernandez-Arjona and Filipović, 2022; Chen et al., 2026).

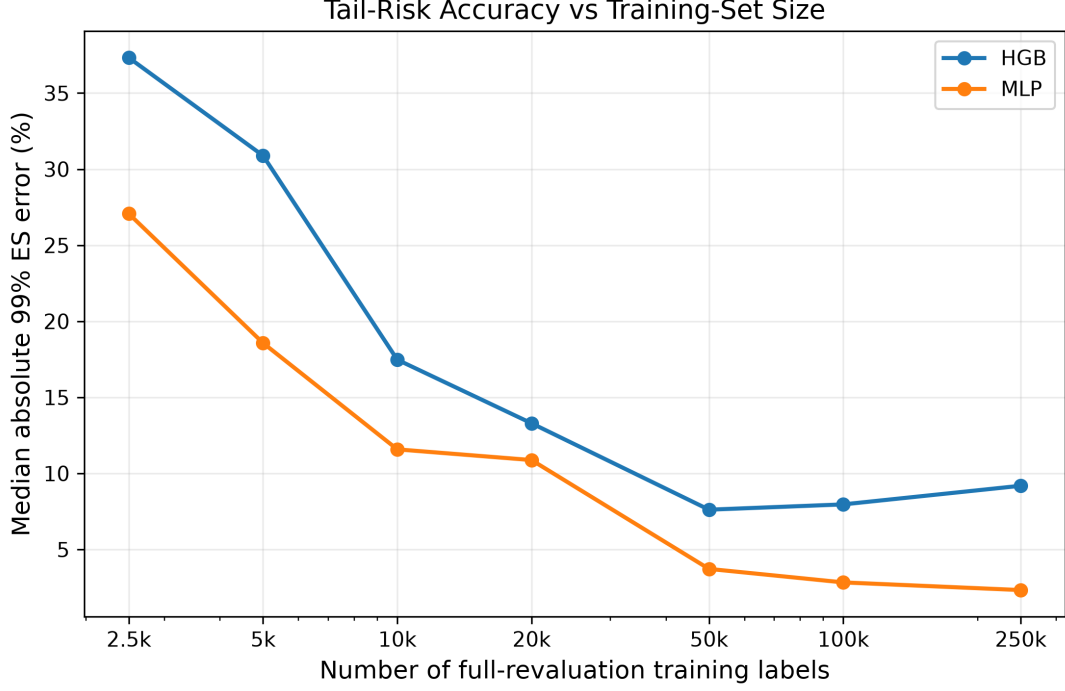


Figure 2: Median overlap with the true worst 1% of scenarios.

3.3 Computational Trade-Off

The median compute-only runtime for direct full revaluation of one million scenarios was 30.01 seconds. Scenario generation alone required 0.515 seconds and Python feature preprocessing required 0.064 seconds. For the 50,000-observation MLP, generating the full-revaluation training labels required 1.51 seconds and median model training required 24.03 seconds. Once trained, a one-million-scenario ML risk calculation required approximately 1.20 seconds including scenario generation, preprocessing and neural-network inference, corresponding to a recurring compute-only speedup of approximately $30.01/1.20 = 25.05$ times.

The surrogate nevertheless incurs an upfront cost because exact full-revaluation labels must first be generated and the network must then be fitted. Let T_{labels} denote label-generation time, T_{train} model-training time, T_{full} the cost of one direct full-revaluation risk run and $T_{\text{ML},\text{recurring}}$ the cost of one subsequent ML-based run. The continuous number of evaluations required to recover the upfront investment is

$$K^* = \frac{T_{\text{labels}} + T_{\text{train}}}{T_{\text{full}} - T_{\text{ML},\text{recurring}}}. \quad (10)$$

The numerator represents the one-off cost of creating the surrogate, while the denominator represents the amount of computation saved on each subsequent one-million-scenario evaluation. For the 50,000-observation model, the combined upfront cost was 25.54 seconds and the first complete ML evaluation required 26.74 seconds, still below the 30.01-second cost of direct full revaluation. The resulting continuous break-even point was approximately 0.89 evaluations.

Table 2: MLP accuracy–computation trade-off.

Labels	ES Error (%)	Upfront Cost (s)	Recurring Cost (s)	Break-Even Run
20,000	10.88	7.83	1.23	1
50,000	3.72	25.54	1.20	1
100,000	2.83	60.40	1.26	3
250,000	2.33	114.36	1.43	5

The 250,000-observation model provides the strongest predictive performance, but the improvement comes with substantially greater upfront cost. The 50,000-observation model therefore provides the strongest practical balance between tail-risk accuracy and computational efficiency in this experiment.

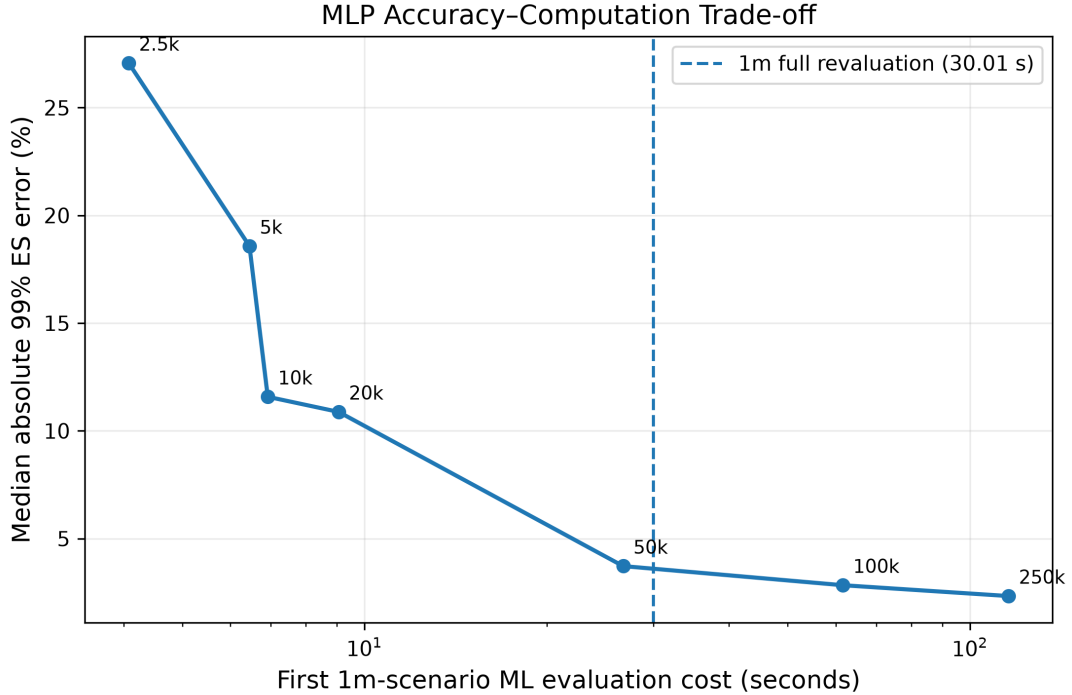


Figure 3: Accuracy–computation trade-off for the MLP.

3.4 Bootstrap Uncertainty

A paired non-parametric bootstrap was used to quantify uncertainty in the difference between the frozen 50,000-observation MLP and full revaluation. Bootstrap resampling provides a general method for estimating sampling distributions from observed data without imposing a specific parametric form (Efron, 1979). Three hundred resamples of size one million were drawn with replacement, and the same scenario indices were applied to both the full-revaluation and MLP losses in every replicate. This pairing removes much of the scenario-sampling variation that would otherwise obscure the difference between the two methods.

For the frozen 50,000-observation MLP, predicted 99% VaR was 1.72% above the full-revaluation value, with a 95% paired bootstrap interval of 1.35% to 2.09%. The model underestimated 99% Expected Shortfall by 1.65%, with a corresponding 95% interval of -2.46% to -0.90% . Since neither interval contains zero, the observed differences are unlikely to be attributable purely to Monte Carlo sampling variation in the benchmark. The ES result is particularly relevant because an underestimate of extreme average losses is anti-conservative.

These confidence intervals are conditional on the specific frozen model and should therefore be distinguished from the five-training-sample robustness experiment, which instead measures variability resulting from different training datasets.

3.5 Scenario Diagnostics

The generated benchmark scenarios were checked against the intended synthetic distribution before drawing conclusions from the surrogate experiments. Empirical means of the eleven risk factors were close to zero, while empirical standard deviations closely matched their target values. The mean empirical spot–spot correlation was 0.3501 compared with a target of 0.35, mean volatility–volatility correlation was 0.5010 compared with 0.50, and mean spot–volatility correlation was -0.2997 compared with -0.30 . The maximum absolute sample correlation between the rate shock and another factor was approximately 0.0032.

The lower volatility floor was also checked because the heavy-tailed Student- t shocks can theoretically generate sufficiently negative volatility changes to make an unfloored volatility non-positive. Only 122 of five million individual volatility observations reached the floor, and only 0.0092% of benchmark scenarios contained at least one floor event. The floor therefore prevents invalid model inputs without materially determining the aggregate results.

4 Discussion

The results indicate that machine-learning surrogates can substantially reduce the computational cost of Monte Carlo tail-risk estimation in this controlled setting, but only when the surrogate has sufficient training data and an architecture suited to the pricing function. A central finding is that conventional regression accuracy alone is not sufficient for evaluating a risk surrogate. A model can perform reasonably well across typical scenarios while making materially larger errors in the small set of extreme scenarios that determines VaR and Expected Shortfall. Tail-specific metrics such as ES error, worst-tail RMSE and tail overlap are therefore necessary when judging whether a surrogate is suitable for risk measurement.

The MLP was substantially better suited to this problem than HGB. Increasing the HGB training sample did not eliminate its performance gap, whereas the neural network continued to improve as more full-revaluation labels became available. This is consistent with other research showing that neural or other machine-learning approximations can be used as fast valuation and risk surrogates (Fernandez-Arjona and Filipović, 2022; Chen et al., 2026). The present results additionally demonstrate that the quantity of expensive full-revaluation training data is itself an important design choice: moving from 20,000 to 50,000 labels produces a large improvement in tail accuracy, whereas moving from 50,000 to 250,000 produces progressively smaller improvements at a much greater upfront computational cost.

The optimal training size therefore depends on how frequently the surrogate will be reused and how much residual risk-measure error is acceptable. A frequently reused model can justify a larger initial training investment, whereas 50,000 observations provide a more attractive compromise in this experiment because the model already achieves strong tail performance and breaks even computationally within the first one-million-scenario evaluation. The bootstrap analysis also shows that a strong surrogate should not be treated as an exact substitute for the original engine: the frozen 50,000-observation network exhibits a small systematic VaR overestimate and ES underestimate. This reinforces the need for explicit tail-focused validation rather than relying solely on conventional predictive metrics.

5 Limitations

The portfolio and scenario model are synthetic, so the numerical thresholds identified in this study should not be interpreted as universal. The portfolio contains only European options and stock positions, and all options are repriced using Black–Scholes–Merton. A real institutional portfolio may contain American, path-dependent, credit or exotic derivatives whose full valuation is substantially more computationally expensive. In such settings the potential benefit of a surrogate may be larger, but the mapping being approximated may also be more difficult. The risk-factor volatilities and correlations used here are controlled assumptions rather than parameters calibrated from historical or option-market data. The Student- t scenario model captures heavy tails and correlated shocks, but does not model stochastic volatility, jumps, liquidity effects, transaction costs or changes in market dependence through time.

The computational comparison is also deliberately compute-only. CSV writing, persistent storage and communication between the C++ and Python components are excluded, meaning that the approximately 25-times recurring speedup should not be interpreted as complete production-system latency. Absolute timing results also depend on hardware and software implementation. Finally, five independent training pools provide useful evidence that the learning curve is not driven by a single favourable sample, but they do not completely characterise training-sample uncertainty. The paired bootstrap addresses a different source of uncertainty—benchmark scenario sampling conditional on one frozen model—and therefore should not be interpreted as incorporating uncertainty associated with retraining the network.

6 Conclusion

This paper investigated whether machine-learning surrogates can reduce the computational cost of Monte Carlo tail-risk estimation while retaining useful accuracy. A C++ full-revaluation engine generated scenario P&L for a nonlinear portfolio of 500 European options, while machine-learning models learned the mapping from eleven simulated market shocks directly to portfolio P&L. The MLP substantially outperformed Histogram Gradient Boosting and improved consistently in its underlying prediction quality as the number of full-revaluation labels increased. Across five independent training samples, 50,000 labels produced median 99% VaR error of 0.69%, median 99% Expected Shortfall error of 3.72% and median worst-1% tail overlap of 89.37%. Increasing the training set to 250,000 observations improved median ES error to 2.33%, tail overlap to 91.98% and R^2 to 0.965, but at a considerably greater upfront cost.

The 50,000-observation MLP reduced the recurring compute cost of a one-million-scenario risk evaluation from 30.01 seconds to approximately 1.20 seconds, corresponding to a compute-only speedup of roughly 25 times, while its initial label-generation and model-training cost was recovered within the first million-scenario evaluation. The evidence therefore supports the central hypothesis that machine-learning surrogates can materially reduce Monte Carlo tail-risk computation in a nonlinear derivatives portfolio. However, the results also show that the answer is conditional on model architecture, training-data quantity and explicit tail-risk validation. In this experiment, approximately 50,000 full-revaluation labels provide the strongest practical balance between computational saving and extreme-tail accuracy.

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